Oleksii Krasnoshtanov

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Education

Breda University of Applied Sciences — Breda, Netherlands
BSc Applied Data Science & Artificial Intelligence; 9.1/10 (4.0 GPA); BUas Scholarship — Sep 2024 – Present

Projects & Technical Work

CV Pipeline – Plant Phenotyping Platform | *5-person group project, BUas* | Code | Docs — Apr – Jun 2026
- Built an MLOps platform for plant organ segmentation and root-tip detection: U-Net (ResNet-34) behind an authenticated FastAPI backend, with a Next.js frontend, PostgreSQL and Prometheus
- Ran the model lifecycle from Airflow DAGs: data versioning, Azure ML training, drift detection on confidence, latency and error rate, and retraining driven by researcher feedback on flagged predictions
- Owned deployment and CI/CD behind an 85% coverage gate and 430 tests: one codebase shipping to Docker Compose, on-premise Portainer and Azure Container Apps, with blue-green cutover, health-gated rollback and OIDC cloud auth

MARBET Event Assistant – On-Premise RAG Chatbot | *2-person sprint* | Code — Apr – May 2025
- Self-hosted RAG assistant answering event questions from the client's confidential documents, in their production: Llama 3.3 70B via Ollama, LangChain, FAISS, multi-format ingestion and a Gradio web interface
- Grounded by construction: answers only from retrieved context, each returned with the source chunks and their relevance, and an explicit fallback when the documents do not cover the question

Measurement Trust Project | *Client project – CEVA Logistics, BUas; team of 4* — Sep 2026 – Jan 2027
- Data engineer on size-group misclassification at bol's BFC1/2 (bol-owned, CEVA-operated), driving fee disputes and re-measurement requests; obtained partly anonymised data from CEVA and bol, liaised with Logistore and third parties, co-wrote BRDs keeping proposals GDPR-compliant, ethical and clear to non-technical stakeholders

Dutch Sign Language Fingerspelling Trainer | *Solo rebuild of a group project* | Code | Docs — May – Sep 2026
- Real-time NGT alphabet trainer: MediaPipe landmarks feed a PyTorch residual MLP for the 24 static letters (95% validation) and a bidirectional LSTM over 30-frame sequences for the two needing motion (92%)
- Sole author: FastAPI WebSocket backend streaming to a Next.js 16 / TypeScript frontend, bilingual EN/NL, containerised with CI and published documentation

NPEC Plant Phenotyping System | *Individual project for NPEC, BUas* | Code — Nov 2025 – Jan 2026
- Ranked **#2 of 77** in a Kaggle root-segmentation competition and selected to present; built the full loop from U-Net segmentation and root-tip extraction to OT-2 pipette control in PyBullet, PID benchmarked against PPO

Hackathons & Competitions

BrabantHack 2026 | *Winner – Demcon Deep Tech Track* | Code — Apr 2026
- Won the Demcon track with a model that locates a pedestrian standing entirely outside the camera frame from the shadow they cast into it: 0.626 mean IoU on the organisers' hidden test set; team of 3
- Drove the gain by re-describing the box relative to the frame edge it hides behind – side, distance, width, height, y-centre – on a ResNet-50 trunk fused with 19 shadow descriptors: 0.468 to 0.610 mean IoU, reproducible from the repository

HERE Technologies Hackathon | *RDM Verification Challenge* — May 2026
- Built an XGBoost classifier for Restricted Driving Manoeuvre verification combining probe GPS features with Mapillary and OSM geospatial signals; designed synthetic data scrambling for NDA-compliant delivery

Technical Skills

Languages: Python, SQL, TypeScript
Generative AI & LLMs: LangChain, Ollama, FAISS, Llama 3.3, RAG, semantic chunking, prompt grounding, Gradio
ML / Deep Learning: PyTorch, Scikit-learn, XGBoost, HuggingFace Transformers, Faster-Whisper, OpenCV
MLOps & Cloud: Azure ML, Azure Container Apps, Airflow, MLflow, Prometheus, Trivy
Infrastructure: pytest, GitHub Actions CI/CD, Docker, Docker Compose, Portainer, uv, Ruff
Backend & Data: FastAPI, Pydantic, PostgreSQL, SQLAlchemy, Alembic, Pandas, NumPy
Frontend: Next.js, React, Tailwind CSS, WebSockets
Spoken Languages: English (fluent), Ukrainian (native), Russian (fluent), Polish (basic)